LOT PAGER - Draft Manual

LOT PAGER — Hardware Computer Connected to lot-systems.com

Doc: LOT-HW-01 · **Session:** 2026-11-08 (S1) · **Owner:** Vadik, Inventor, COSMO® CIA **Status:** PLAN + BOM + ROADMAP. No hardware ordered. All prices are planning estimates (see §9).

0. TL;DR

- Device: 40 × 40 × ~6 mm stainless-steel pager. Back = polished mirror steel. Front = small screen, camera, one button (“Copy”).
- Brain: ESP32-S3 (Wi-Fi/BLE, camera + display capable). Sensors: BME688 (weather + gas), IMU, ambient light.
- Link to site: new `/api/device/*` endpoints on LOT API. Site pushes short notices (“Coffee time!”) to the screen. Button press writes a `device_copy` event into the Log tab.
- Build: PCBWay (PCB + SMT + CNC steel), pilot 10 units, then 100.
- Estimated landed cost: **~$95–130/unit, ~$10–13k for 100** (estimate).
- **Three physics risks need a decision before PCB layout** (§3): 5 mm thickness, steel blocking Qi charging, steel blocking Wi-Fi.

1. Requirement map (your 19 points)

#	Requirement	Where addressed
1	PCBWay	§5 supplier plan
2	Pager-like notification from AI site	§4 architecture, LOT-HW-03
3	2-part stainless body	§3, §5
4	Flat silver square 4×4 cm × 5 mm	§3 (risk R1: 5 mm is very tight; fallback 6.5 mm)
5	Camera	§2 BOM, §3
6	LOT API connector	LOT-HW-03
7	PDF manuals	LOT-HW-06 + PDF in this folder
8	Compress info each session	§8 session protocol (≤1 page digest, ≤80-char payloads)
9	Firmware documents	LOT-HW-04
10	Software to connect to firmware	LOT-HW-05 (pairing/flash tool)
11	Separate documents	LOT-HW-01…06 + BOM CSV
12	Charger	§2 (Qi puck, see R2)
13	100-unit run	§6 roadmap, §7 cost
14	Weather sensor	BME688
15	AI-grade off-the-shelf sensors	BME688 (on-chip gas AI/BSEC), BMI270 IMU (gesture), LTR-559 light

#	Requirement	Where addressed
16	Button “Copy” → Log tab	LOT-HW-03 (`device_copy` event)
17	One side polished stainless	back plate, mirror polish #8
18	Other side camera + screen + button	front plate
19	Wireless charger	Qi, see R2

2. Component buying list

Links go to vendor/search pages, not verified live listings (the routine’s network could not reach vendors). Verify part numbers at purchase. Full table: `LOT-HW-02-BOM.csv`.

Block	Part	Why	Source
MCU	ESP32-S3FH4R2 (or module ESP32-S3-MINI-1U w/ external antenna)	Wi-Fi/BLE, DVP camera, SPI display, deep sleep	https://www.lcsc.com/search?q=ESP32-S3FH4R2
Camera	OV2640 FPC module (small 8×8 mm, 2MP)	cheap, ESP32 supported	https://www.aliexpress.com/wholesale?SearchText=OV2640+fpc
Screen	0.96” 160×80 IPS SPI (ST7735/ST7160) or 0.91” 128×32 OLED	simple text notices, thin	https://www.lcsc.com/search?q=0.96+inch+IPS+SPI
Weather	Bosch BME688	temp/humidity/pressure/gas, on-chip AI (BSEC)	https://www.digikey.com/en/products/result?keywords=BME688
IMU	Bosch BMI270	tap/gesture/step, low power	https://www.digikey.com/en/products/result?keywords=BMI270
Light	LTR-559 / VEML7700	auto-dim screen	https://www.lcsc.com/search?q=VEML7700
Haptic	6 mm LRA + DRV2605L driver	pager buzz (silent)	https://www.digikey.com/en/products/result?keywords=DRV2605L
Button	Ultra-low-profile tact switch, ≤1.5 mm	“Copy”	https://www.lcsc.com/search?q=SMD+tact+switch+1.5mm
Battery	LiPo ~3×20×25 mm, 120–150 mAh w/ protection	fits thin stack	https://www.digikey.com/en/products/result?keywords=lipo+301525
PMIC	BQ25100 or MCP73831 + 3.3 V LDO/buck (TPS62740)	charge + power	https://www.digikey.com/en/products/result?keywords=BQ25100
Wireless RX	Qi receiver IC (e.g. SGM-/CPS4041 class) + 0.3 mm thin RX coil with ferrite	wireless charge	https://www.lcsc.com/search?q=qi+receiver+coil
Charger (dock)	5 W Qi puck, off-the-shelf, ×10 pilot / ×100	item 12 / 19	https://www.aliexpress.com/wholesale?SearchText=qi+wireless+charger+5w
USB-C pogo/	4-pin pogo pads on PCB	flashing + fallback charge	on PCB

Block	Part	Why	Source
test			
Body	316L stainless, 2 CNC parts, mirror-polished back	items 3,4,17	https://www.pcbway.com/cnc-machining-quote.html
PCB + SMT	4-layer, 0.8 mm, ENIG	item 1	https://www.pcbway.com/

3. Physics risks (decide before layout)

R1 — 5 mm thickness. Stack: front steel 0.8 + display 1.6 + PCB 0.8 + battery 3.0 + back steel 0.8 = 7.0 mm before coil/camera. 5 mm is not achievable with camera + display + battery + Qi. Options: (A) 6.5–7 mm body (recommended); (B) 5 mm "shell" with no camera/Qi. **R2 — Steel blocks Qi.** Eddy currents in stainless kill wireless charging. Fix: non-metal window behind the coil (ceramic/glass/PEEK insert in the back or a ceramic ring), which breaks the "fully polished" back. Alternative: charge through the front via a plastic-glass window and keep the back pure mirror. Fallback: pogo-pin dock. **R3 — Steel blocks RF.** Wi-Fi/BLE antenna needs a non-metal slot: a plastic/ceramic gasket in the seam between the two parts (the gasket doubles as waterproof seal). Use u.FL/ceramic antenna at the gasket edge. **R4 — Camera/screen windows.** Sapphire or glass 0.5 mm inlays in front plate; adhesive-sealed. **R5 — Privacy.** Camera must have a hardware shutter or LED + off-by-default; log capture events.

4. Architecture

```
lot-systems.com (Fastify API) ──SSE/WebSocket──▶ LOT PAGER (ESP32-S3)
  AI engine → notice (≤80 chars)        ◀──HTTPS POST── button "Copy", sensor, photo
  Log tab  ◀── event: device_copy
```

Existing repo hooks: `src/server/routes/api.ts` has `/sync` SSE, `/weather`, `/logs`, `/live-message`; `Log` model has `event`, `text`, `metadata`. Auth today is a cookie JWT, so the device needs its own token (LOT-HW-03).

5. Supplier plan (PCBWay)

1. PCB fab + SMT assembly (turnkey): Gerbers, BOM, pick&place.
2. CNC 316L, 2 parts, polish #8 mirror on back; ask for DFM review + sample.
3. Housing gasket (silicone/PEEK) via PCBWay 3D print/CNC.
4. Pilot 5–10 → EVT, then 100 → DVT/PVT. Request quote at each stage; PCBWay lead times are quote-dependent.

6. Roadmap (est. 22–26 weeks)

Phase	Weeks	Output	Gate
P0 Decide	0–1	Resolve R1–R3, lock thickness	Vadik sign-off
P1 Proto on dev kit	1–5	ESP32-S3 devkit + screen + BME688 talks to API; server endpoints live	Notice shown on screen, Copy → Log
P2 Schematic/PCB	4–9	v0.1 schematic, layout, DFM	PCBWay DFM pass

Phase	Weeks	Output	Gate
P3 Body CAD	6–11	2-part steel, gasket, windows	CNC sample quote
P4 EVT (10 units)	10–16	Assembled units, charging proven through window	Qi ≥ 70 % eff., RF range ≥ 10 m
P5 Firmware/Software	8–18	OTA, pairing, flasher tool, manuals	All HW docs v1.0
P6 DVT/PVT	16–22	20-unit run, drop/seal/thermal tests	Pass criteria
P7 Production 100	22–26	100 units, QA, packaging	Ship

7. Cost estimate (100 units, USD, planning only)

Item	/unit	×100
PCB + SMT	14–20	1.4–2.0k
Components (MCU, camera, display, sensors, battery, Qi)	22–32	2.2–3.2k
CNC stainless pair + polish + gasket + windows	45–65	4.5–6.5k
Qi dock	5–8	0.5–0.8k
Pack, test jig, QA, shipping	6–10	0.6–1.0k
Total	**~92–135**	**~9.2–13.5k**

Plus one-off: stencils/fixtures, EVT/DVT iterations (~$1.5–3k), certification FCC/CE if sold (~$5–15k, not included).

8. Session protocol (item 8)

Each working session ends with one MD report in `docs/hardware/sessions/`: ≤1 page — decisions, open risks, next 3 actions. Device-bound text: ≤80 chars, one idea, no emoji (MILITARY PURITY).

9. Honest limits of this session

- No network access to lot-systems.com / brand / institute pages (blocked), so brand tone was taken from repo docs (military purity, no emoji). Re-run with those domains allowed to align the design language.
- Prices, lead times and part availability are estimates, not quotes.
- No PDF manuals for end users can be final until R1–R3 are decided; a draft PDF is included.

10. Next 3 actions

1. Decide thickness (6.5–7 mm) and charging window approach (R1/R2).
2. Implement `/api/device/*` (LOT-HW-03) and test with an ESP32-S3 devkit.
3. Request PCBWay CNC + PCB quotes using LOT-HW-02. # LOT-HW-03 — LOT API Device Connector (proposed) Status: SPEC. Not yet implemented in `src/server`. Fits existing Fastify routes (`api.ts`) and `Log` model (`event`, `text`, `metadata`, `context`).

Auth

Existing auth is cookie JWT (user session). Device uses a per-device bearer token minted in Settings ("Pair device"), stored hashed server-side, scoped to `device:*`, revocable. TLS only.

Endpoints

Method	Path	Purpose
POST	`/api/device/pair`	Exchange 6-digit code (shown in site) for device token
GET	`/api/device/stream`	SSE: `notice`, `config`, `ping` events to device
POST	`/api/device/event`	Device → site: `copy`, `sensor`, `battery`, `photo_meta`
POST	`/api/device/photo`	Optional JPEG upload (off by default)
POST	`/api/admin/device/notice`	Site/AI sends a notice to a user's device

Notice payload (site → device)

```
        {"id":"n_01","text":"Coffee
time!","ttl_s":600,"haptic":"double","ts":1762600000}
```

`text` max 80 chars, ASCII + basic Latin. Device acks with `event: ack`.

Button "Copy" (device → site)

```
        POST /api/device/event
        {"type":"copy","notice_id":"n_01","ts":1762600042,"battery":0.81}
```

Server writes `Log.create({userId, event:'device_copy', text:'Coffee time!', metadata: {deviceId, noticeId}})`. Add `'device_copy'` to `displayableEvents` in `GET /logs` so it appears in the Log tab.

Sensor event

`{"type":"sensor","t_c":21.4,"rh":44,"hpa":1013.2,"gas_ohm":81234,"iaq":62}` → stored as `device_sensor` (not displayed by default; feeds QIE environment signal).

AI notification flow

Scheduled job / AI engine decides a nudge → trims to ≤80 chars → `POST /api/admin/device/notice` → SSE push → screen + haptic. Rate limit: 1 notice / 5 min default; quiet hours from user settings.

Offline

Device keeps last 5 notices; queues up to 50 events in flash; syncs on reconnect. # LOT-HW-04 — Firmware Document Target: ESP32-S3, ESP-IDF 5.x (C/C++), FreeRTOS. ## Modules `net` (Wi-Fi provisioning, TLS, SSE client) · `api` (LOT connector, token store in NVS) · `ui` (text screen, 1 notice + status bar) · `input` (button: short=Copy, long=dismiss/pair) · `sensors` (BME688 w/ BSEC, BMI270,

light) · `haptic` (DRV2605L patterns) · `camera` (off by default; shutter on request) · `power` (deep sleep, charge state) · `ota` (signed images, A/B partitions). ## Power model Deep sleep between 30 s SSE keepalive wakes is not feasible with persistent SSE; use light sleep + Wi-Fi power-save (DTIM) target < 3 mA avg; fallback poll every 60 s for ~7 days on 130 mAh (to be measured in EVT). ## States BOOT → PROVISION (BLE/pogo) → PAIR (code) → IDLE → NOTICE → ACK → IDLE; CHARGING overlay. ## Security Secure boot + flash encryption on production units; token in encrypted NVS; OTA signature check. ## Tests Unit (host), HIL button/screen, 72 h soak, drop, thermal (battery < 45 °C). # LOT-HW-05 — Companion Software (flash + pair) Web tool (Web Serial / esptool-js) hosted under lot-systems.com: flash firmware over pogo/USB-C, write Wi-Fi creds, show pairing code, run self-test (screen, button, sensors, haptic, charge), print result for QA log. Factory mode: batch of 100, scan serial → write device ID → store pass/fail CSV. # LOT-HW-06 — Manuals (PDF set) 1. Quick Start (1 page): charge, pair, first notice, Copy button. 2. User Manual: notices, haptics, quiet hours, camera privacy, care of polished steel. 3. Safety & Battery sheet (LiPo, charging, disposal). 4. Service/Assembly Manual (factory): torque, gasket, window adhesive, test jig. 5. Developer Guide: API, firmware build, OTA. Style: LOT military purity — no decoration, no emoji.